

A20.

(1) 设 A 为选甲盒, B 为取到 1 个红球 1 个白球

$$\begin{aligned} P(B) &= P(A) \cdot P(B|A) + P(\bar{A}) \cdot P(B|\bar{A}) \\ &= \frac{1}{2} \times \frac{C_3^2}{C_5^2} + \frac{1}{2} \times \frac{C_2^2}{C_6^2} = \frac{11}{60} \end{aligned}$$

(2) 设 C 为从甲盒取到红球, D 为从乙盒中取到红球

$$\begin{aligned} P(D) &= P(C) \cdot P(D|C) + P(\bar{C}) \cdot P(D|\bar{C}) \\ &= \frac{3}{5} \times \frac{3}{7} + \frac{2}{5} \times \frac{2}{7} = \frac{13}{35} \end{aligned}$$

(3) 设 F 为取到一个红球一个白球.

$$P(F) = P(C) \cdot P(F|C) + P(\bar{C}) \cdot P(F|\bar{C})$$



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$$= \frac{3}{5} \times \frac{C_3^1 C_4^1}{C_7^2} + \frac{2}{5} \times \frac{C_2^1 C_5^1}{C_7^2} = \frac{56}{105}$$

A23.

(1) 设 A 为取到调试好的枪, B 为命中

$$P(B) = P(A) \cdot P(B|A) + P(\bar{A}) \cdot P(B|\bar{A})$$

$$= \frac{3}{4} \times 0.6 + \frac{1}{4} \times 0.05 = 0.4625$$

$$(2) P(\bar{A}|B) = \frac{P(\bar{A}B)}{P(B)} = \frac{P(\bar{A}) \cdot P(B|\bar{A})}{P(B)} = \frac{1}{37}$$

A25.

(1) 设 A_1, A_2, A_3 分别为不吸烟, 每天 1~10 支, 每天吸烟 10 支事件. $P(A_1) = 0.4$, $P(A_2) = 0.6 \times 0.5 = 0.18$, $P(A_3) = 0.42$

B 为先天畸形儿

$$P(B) = (P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3))$$

$$= 0.4 \times 0.18\% + 0.18 \times 1.4\% + 0.42 \times 2.1\%$$

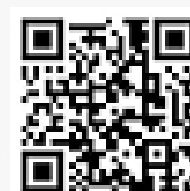
$$= 0.01454$$

$$(2) P(A_3|B) = \frac{P(A_3B)}{P(B)} = \frac{P(A_3)P(B|A_3)}{P(B)} = \frac{44}{727}$$

A33

(1) 设 A, B, C 分别为灯管使用寿命大于 1000h, 2000h, 4000h.

$$P(C|A) = \frac{P(CA)}{P(A)} = \frac{P(C)}{P(A)} = \frac{1}{19}$$



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(2) 设 D_1, D_2, D_3 为 0 个、1 个、2 个损坏, D 为至少三个损坏

$$P(D) = 1 - P(\bar{D}) = 1 - P(D_1) - P(D_2) - P(D_3)$$

$$P(D_1) = 0.3^{10}$$

$$P(D_2) = C_{10}^1 \cdot 0.7^1 \cdot 0.3^9$$

$$P(D_3) = C_{10}^2 \cdot 0.7^2 \cdot 0.3^8$$

$$\Rightarrow P(D) = 0.9984$$

A34

设 A 为扔 n 次下至少一次命中目标物

$$P(A) = 1 - P(\bar{A}) = 1 - 0.95^n > 0.5$$

$$\Rightarrow n > 13.5134$$

$\Rightarrow$ 至少扔 14 次

B6

(1) 设 A 为选到第一组 B 为第一次取到优质品

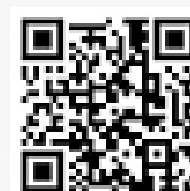
$$P(B) = P(A) \cdot P(B|A) + P(\bar{A}) \cdot P(B|\bar{A})$$

$$= \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{3}{4} = \frac{13}{24}$$

(2) 设 C 为第二次取到的不是优质品

$$P(C|B) = \frac{P(BC)}{P(B)}$$

$$P(BC) = \frac{1}{2} \times \frac{1}{3} \times \frac{20}{24} + \frac{1}{2} \times \frac{3}{4} \times \frac{5}{19} = \frac{2825}{13224}$$



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$$\Rightarrow P(C|B) = \frac{2875}{7163}.$$

B8.

(1) 设 A 为出现雾霾, B 为戴口罩

$$P(B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A})$$

$$= 0.4 \times 0.2 + 0.6 \times 0.01$$

$$= 0.086$$

(2) C 为至少有一位居民戴口罩

$$P(C) = 1 - P(\bar{C}) = 1 - 0.914^3 = 0.236448056$$



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